An Overview Of Wireless Optical Communication

Wireless optical communication (WOC) uses optical carrier in the near-infrared (IR) and visible bands and is considered a viable solution for achieving high-speed and large-capacity communication links

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The arena of wireless communication is mostly dominated by radiofrequency (RF)-based technologies. The constant growth in Internet data traffic is challenging the existing technologies to provide higher throughput, better security, and lower cost. Along with RF, wireless communication also includes the systems that use the infrared (IR) regime of the electromagnetic spectrum. These systems are commonly coined as wireless optical communication (WOC) systems. The use of light in communication was ignited by the demonstration of the first working laser at Hughes Research Laboratories, California, in 1960. With the significant development in optoelectronic components and the market ecosystem over the last few years, WOC has gone through tremendous development.

Indoor and outdoor wireless optical communication

As a broad classification, WOC can be divided into two categories: Indoor WOC and outdoor WOC, commonly known as free-space optical (FSO) communication.

The sub-division of the indoor WOC system is based on the viewing angle of the transmitter and receiver.

Directed LOS system. In this type of WOC system, both the beam angle of the transmitter and field of view of the receiver are very narrow, allowing point-to-point or line-of-sight (LOS) communication. The alignment of the receiver and the transmitter is very crucial to achieve the quality of service. Blocking in the LOS path breaks the communication.

Non-directed LOS system. This type of indoor system is suitable for point-to-multipoint communication. The beam angle of the transmitter and field of view of the receiver are wide enough to avoid any stringent alignment requirement. This system allows limited user mobility.

Diffused system. In this type of system, the transmitter is faced towards the ceiling. The light is reflected from the ceiling and walls multiple times to reach the receiver having a wide field of view. Due to multiple paths from the trans-

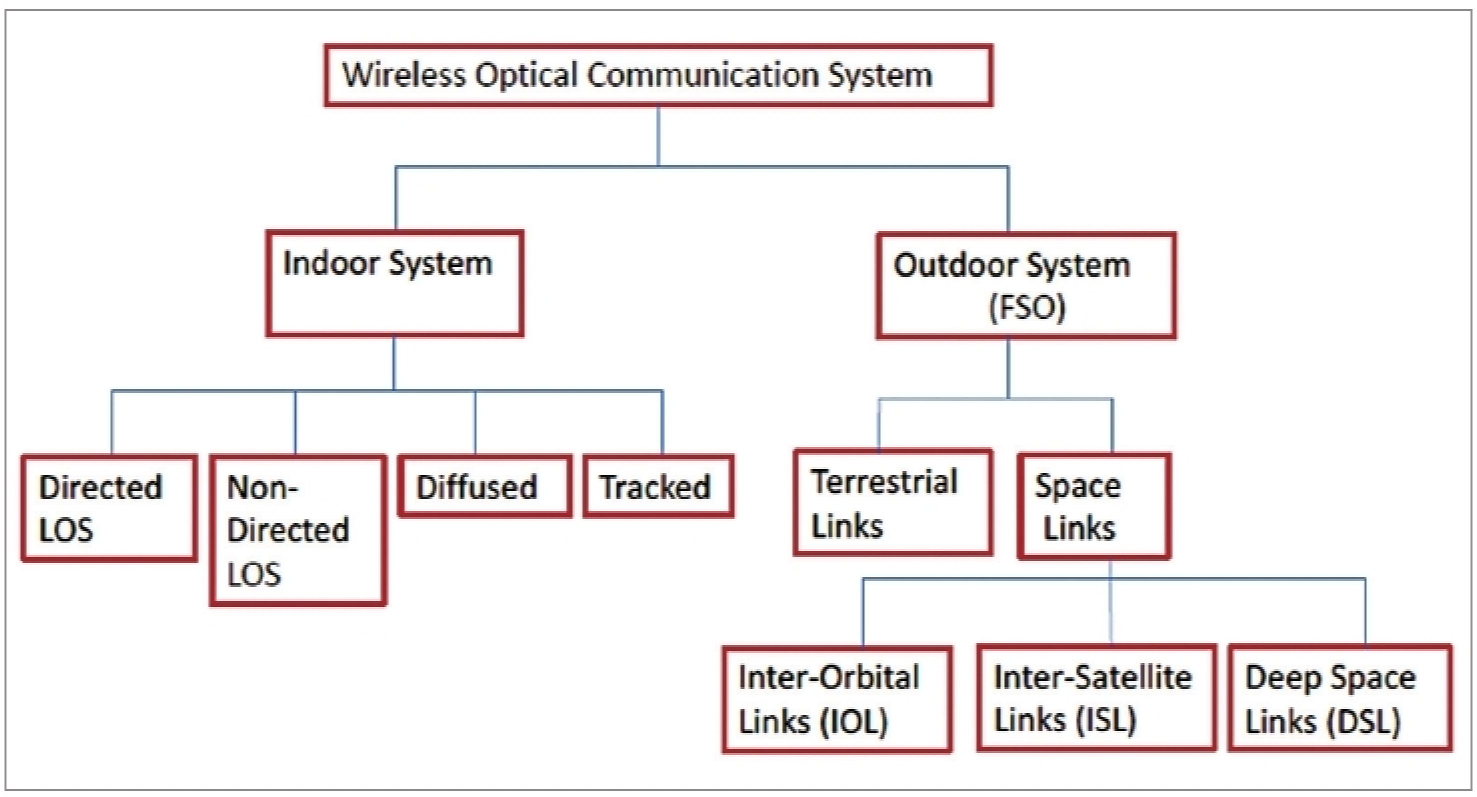

Fig. 1: Classification of WOC system (Source: https://arxiv.org)